

Problem

Avocado has been consumed in recent years due to its nutritional values and popularity. Unlike its high consumption, it is still considered an expensive fruit and not so accessible for everyone and every season. It is intended to explore the price and demand relation for avocado and other factors.

Method

Python is used for correlation between price and demand, price elasticity and optimal price analysis, and data visualisation. For these analyses and visualisations the following libraries are benefited from: pandas, statsmodels.api, matplotlib.pyplot, seaborn, numpy. For further data visualisations and geographical mappings, it is worked with Tableau, the related link can be found in this [link](#).

Insights

According to the regression analysis which results in the price elasticity as $-4,688$, the demand for avocado decreases as the price increases.

The correlation between price and demand is found $-0,19$, which emphasizes the fact that demand slightly decreases as the price increases.

The negative correlation between price and demand can also be seen in Price vs. Demand graph.

The optimal price to maximize the revenue is found 1,159 USD.

Recommendations

The average revenue drops after the optimal price, 1,159 USD, even though the average price goes up due to the decrease in the demand. Hence, it is not recommended to exceed the price of 1,159 USD.

The typical strategy for normal goods, price sensitive products, to decrease the price to increase revenue seems valid for this case.

Optimal Price Effect

